Electronic System Specification

EDC17CV41
Genset Application
NEF and CURSOR
Not Emissioned Engines

Revision 1.4

Revision History

Rev	Date	Description	Authors	Approval
1.0	11.07.14	First issue	G. F. Di Graziano	P. Gamba G. Zottolo
1.1	26.09.14	Updated pin list with MSS, PTO, Cold Start LED	G. F. Di Graziano	P. Gamba G. Zottolo
1.2	21.01.15	Updated para 4, 5, 7, for new Engine Interface Box introduction, MSS input circuit. Updated Appendix D for PTO signals remove	E. Zoncu	V. Palumbo
1.3	21.01.15	Added information on TSC1 Rolling Counter/Checksum use	E. Zoncu	V. Palumbo
1.3.1	30.01.2015	Improved layout	G. F. Di Graziano	V. Palumbo
1.4	06.07.2015	TSC1-VE replaced by TSC1-PE; Added notes on TSC1s time out errors.	G. F. Di Graziano	V. Palumbo

Index

1. ABBREVIATIONS 4
2. REFERENCE DOCUMENTS 5
2.1 Project Specific Documents 5
2.2 Standards Documents 5
2.2.1 SAE 5
2.2.2 ISO 5
3. ENGINE MANAGEMENT SYSTEM 6
4. ARCHITECTURE 9
5. ENGINE CONTROL MODULE 10
5.1 Overview 10
5.1.1 Electrical Characteristics 10
5.1.2 Temperature limitations 16
5.1.3 Handling 18
5.1.4 Type of protection 18
5.1.5 Coating of the Electronic Control Unit 18
5.2 Wiring Harness Architecture 18
5.3 ECM Communication 20
5.3.1 Standard CAN line 22
5.3.2 Special purpose CAN line (FPT ONLY) 23
6. ELECTRICAL BALANCE FOR FPT COMPONENTS 24
7. VEHICLE FUNCTIONS 25
7.1 Engine Torque and Speed Management 25
7.1.1 Multiple state switch 26
7.2 High-Idle setting 27
7.3 Water in Fuel Switch 28
7.4 Engine Electrical heaters 28
7.4.1 Fuel Filter and pre-filter Heating 28
7.4.2 Grid Heater 28
7.5 Lamps, Alarms and gauges 29
7.5.1 Engine speed output 29
7.5.2 High Engine Coolant Temperature Lamp 29
7.5.3 Low Engine Oil Pressure Lamp 29
7.5.4 High Engine Oil Temperature Lamp 29
7.5.5 Diagnostic Lamp 29
7.5.6 Cold Start Lamp 29
8. DIAGNOSIS 31
8.1 On board diagnosis 31
8.2 Off board diagnosis 31

1. ABBREVIATIONS

Abbreviation	Used for
BDS	Battery Disconnection Switch
CAN	Controller Area Network
DTC	Diagnostic Trouble Code
ECM	Engine Control Module
ECU	Electronic Control Unit
EDC17CV41	BOSCH Electronic Diesel Control Unit for NEF and Cursor engines
EOL	End Of Line
PTO	Power Take Off
TBC	To be confirmed
TBD	To be defined
TBU	To be updated

2. REFERENCE DOCUMENTS

2.1 Project Specific Documents

[A.1] CANSpec_Genset_Not_Emissioned v.1.1
[A.2] FPT electrical schematic diagrams: TBD

2.2 Standards Documents

2.2.1 SAE

[B.1] SAE J1939/11 CAN Physical Layer
[B.2] SAE J1939/21 CAN Data Link Layer
[B.3] SAE J1939/71 CAN Vehicle Application Layer
[B.4] SAE J1939/73 CAN Diagnostic Application Layer

2.2.2 ISO

[C.1] ISO 14229-1 UDS – Specifications and Requirements
[C.2] ISO 15765-1 Diagnostic on CAN – General Information
[C.3] ISO 15765-2 Diagnostic on CAN – Network Layer Services
[C.4] ISO 15765-3 Diagnostic on CAN – Application Layer Services
[C.5] ISO 11898 CAN Physical and Data Link Specifications

3. ENGINE MANAGEMENT SYSTEM

The main parts of the Engine Management System are:

- Engine Control Module (ECM)
- Fuel Injection System (FIS)
- Electrical and electronic components (sensors and actuators)

Figure 1 is referred to NEF 4 cylinder Not Emissioned engine. It is valid also for 6 cylinders engine, except only for number of cylinders and injectors.

Figure 2 is referred to Cursor 9 and 13 Not Emissioned engines.

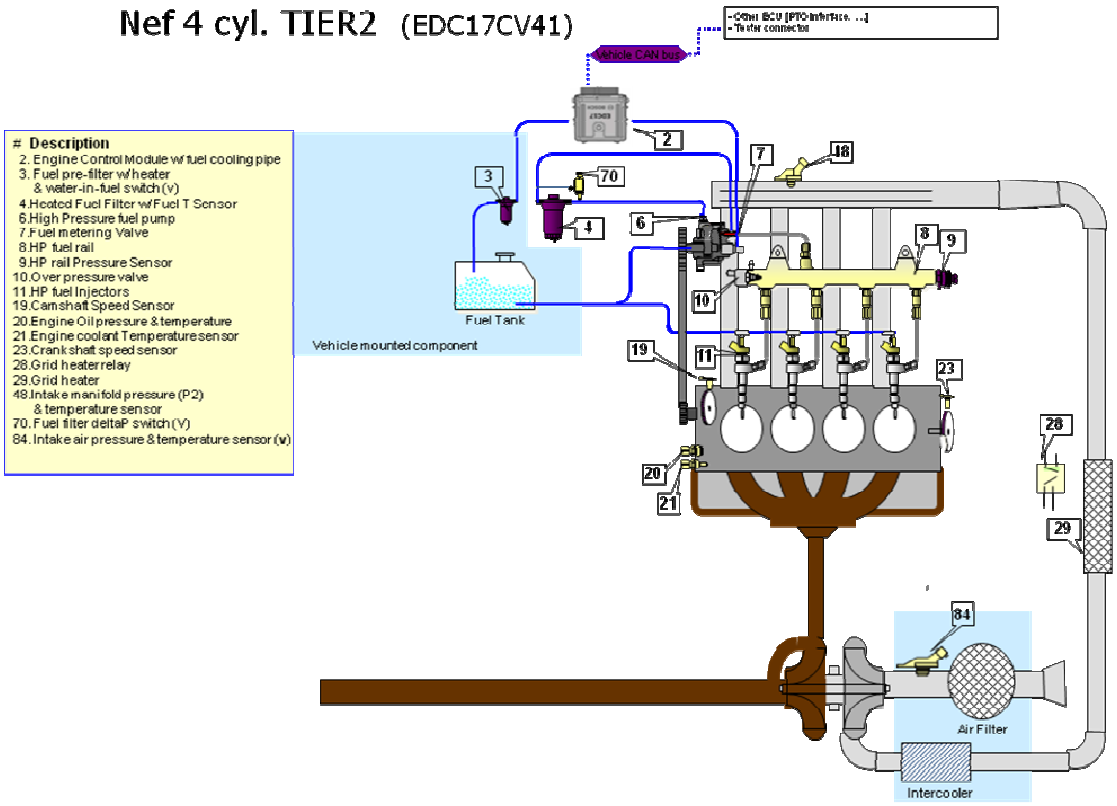


Figure 1 – EMS scheme for NEF 4 cylinders Not Emissioned.

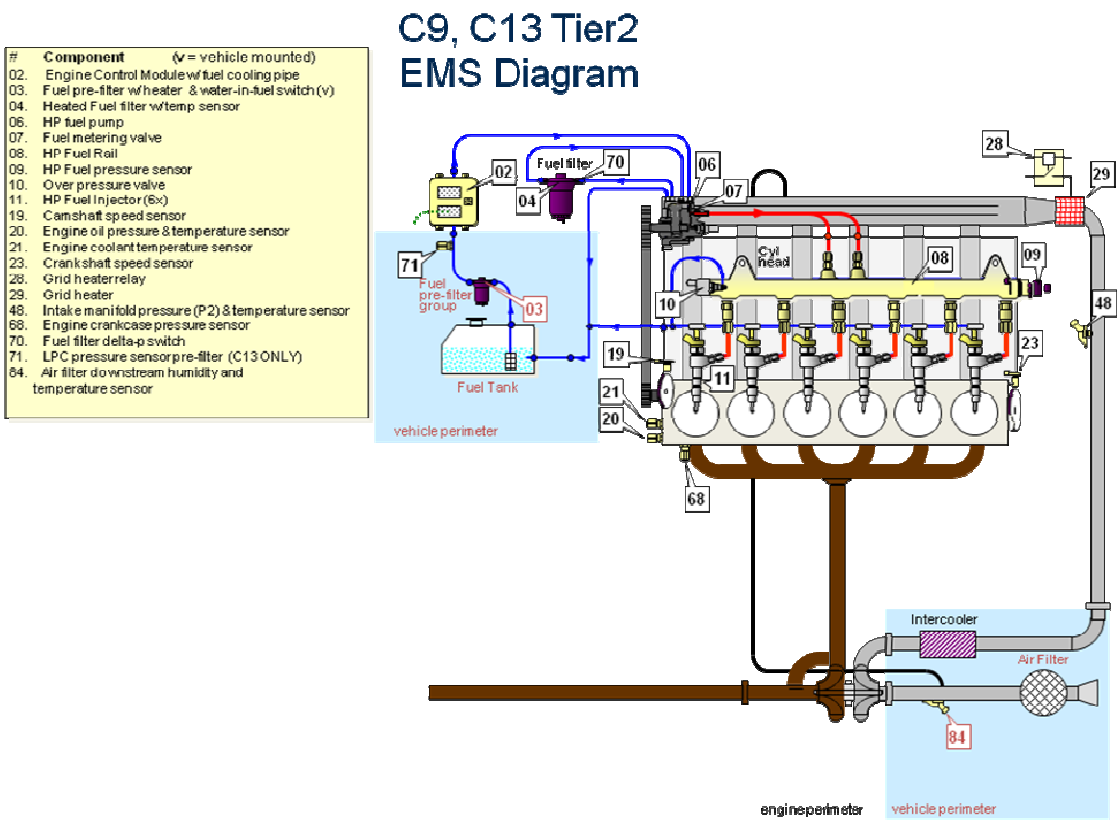


Figure 2 – EMS scheme for Cursor 9, 13 Not Emissioned

4. ARCHITECTURE

In the Figure 3 is shown the connection between EDC17CV41 and the other electrical and electronic devices installed on the power generator.

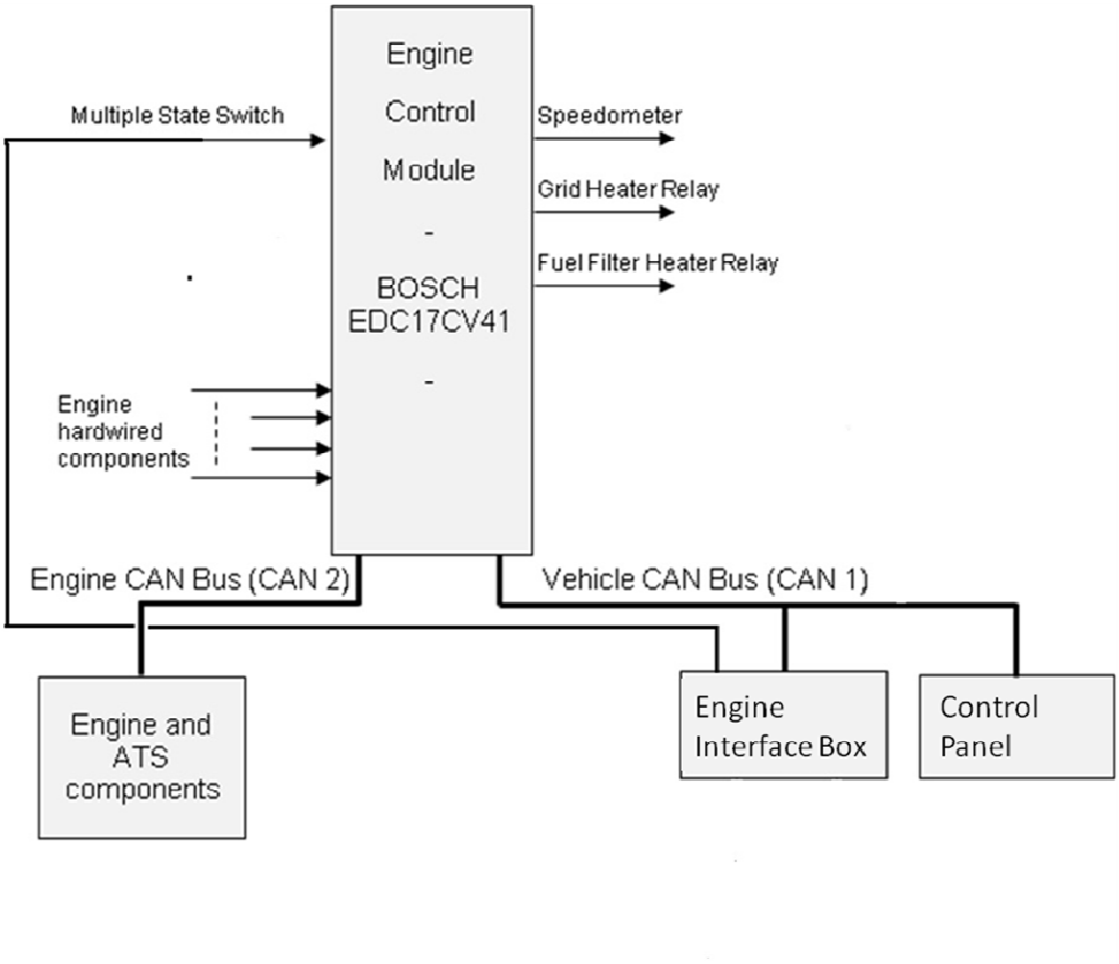


Figure 3 - Main connections between EDC17CV41 and the other electronic devices

For further details on the Engine Interface Box see paragraph 5.3.

5. ENGINE CONTROL MODULE

5.1 Overview

The main features of the Engine Control Unit are:

- Hardware: BOSCH EDC17CV41.
- Supply Voltage: used for 12/24 nominal voltage applications .
- Communication Interfaces: High-Speed CAN.
- Communication Protocols: SAE J1939 for engine control, ISO14229 (UDS) for engine diagnosis.
- Emission level software management: Not Emissioned.

Following are further information about the Electrical and Temperature limitations.

5.1.1 Electrical Characteristics

- Nominal voltage:
  - V_{BAT+} : 12 / 24 V
  - V_{BAT-} : connector pins: 0 V

- Supply voltage range for ECM operation:
  - V_{BAT+} functional range: 9 – 32 V

- Supply nominal voltage:
  - 12V System V_{BAT+} 14 +/- 0.5 V
  - 24V System V_{BAT+} 28 +/- 0.5 V

- Supply ripple:
  - Max value: 2 A_{rms} @ 32 V
  - Min value: 1,5 A_{rms}

- Supply current in Key-Off condition (post after-run phase), measured at room temperature:

  - $I_{ECM, K15=OFF}$: I < 500 µA measured at V_{BAT+} power supply pins.
  - In Key-Off condition all the electrical loads, managed by ECM relay, are disconnected.

- Battery wiring connection:

Recommended value resistance of the wiring harness, between the V_{BAT+}/ECM/V_{BAT-} is ≤ 80mΩ

- Battery reversal protection:
  The ECM has a reverse polarity protection within Supply voltage range 9-32V limits.
  This protection is provided by an internal circuit. During the duration when the supply voltage is applied in reverse polarity to the ECM, the fuse of the power supply lines will not be destroyed.

- ECM operation without battery: Not allowed.

- The system cannot be powered on permanently without engine running.

- Battery Disconnection Switch (if required):
  The Battery Disconnection Switch (BDS) must be connected between the Battery+ pole and the key switched supply.
  The battery must remain connected after Key-Off at least for the whole after-run time duration (180 s). During after-run phase it is prohibited to disconnect the battery. Never disconnect battery with engine running. This operation may cause data loss in the ECM memory.
  There are three allowed BDS electrical configurations:

  - The BDS is connected between the Battery+ pole and the key switched supply, but the ECM positive supply is directly connected to the Battery+ pole. ECM is permanently powered, even after power latch is elapsed (see Figure 4):

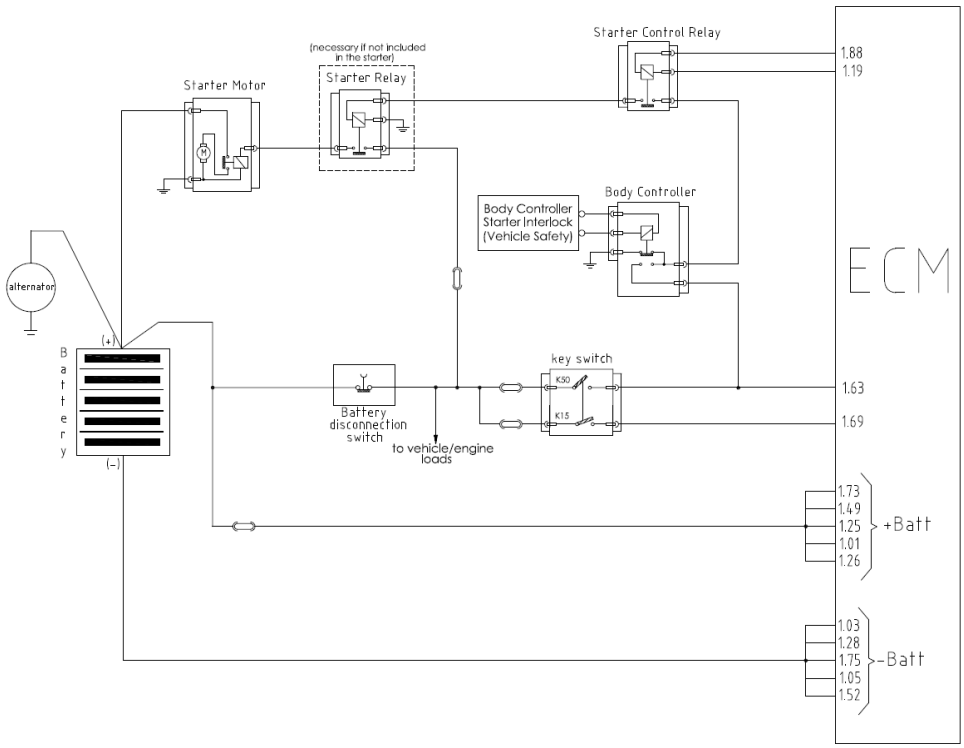


Figure 4 - Battery Disconnection Switch 1st configuration

- The BDS is connected between the Battery+ pole and the ECM positive supply, but it is controlled by external logic, for example by Smart Fuse Box or Vehicle Control Module, that MUST supply the ECM for 180 seconds after BDS opening or key OFF (see Figure 5):

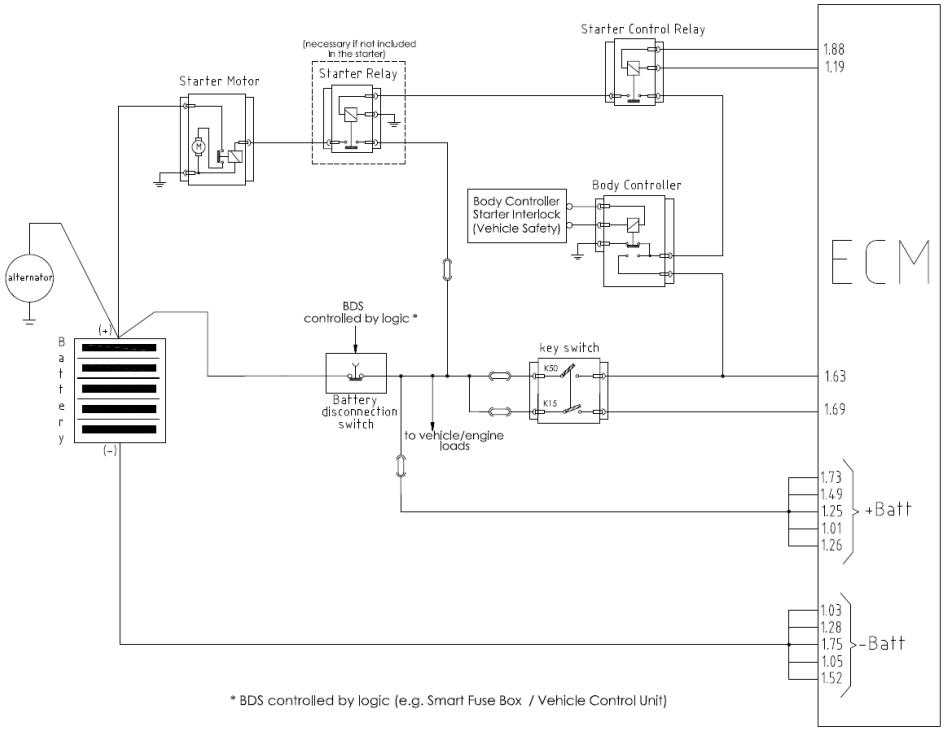


Figure 5 - Battery Disconnection Switch 2nd configuration

- A Time Relay with a delay timer of 180 seconds is connected between the Battery+ pole and the ECM positive supply. The Time Relay MUST guarantee the ECM supply for 180 seconds after BDS opening or key OFF (see Figure 6):

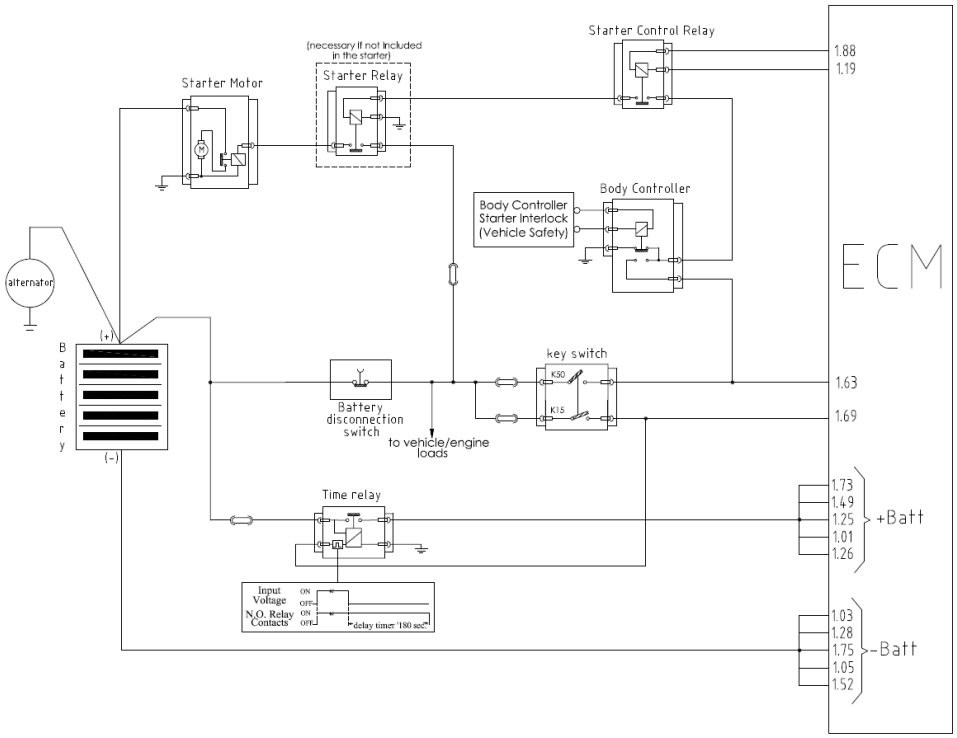


Figure 6 - Battery Disconnection Switch 3rd configuration

- Recommended ECM Ground paths and allowed/forbidden power supply connection set-ups.
    - The following ECM power supply/GND connection set-up is recommended (see Figure 7):

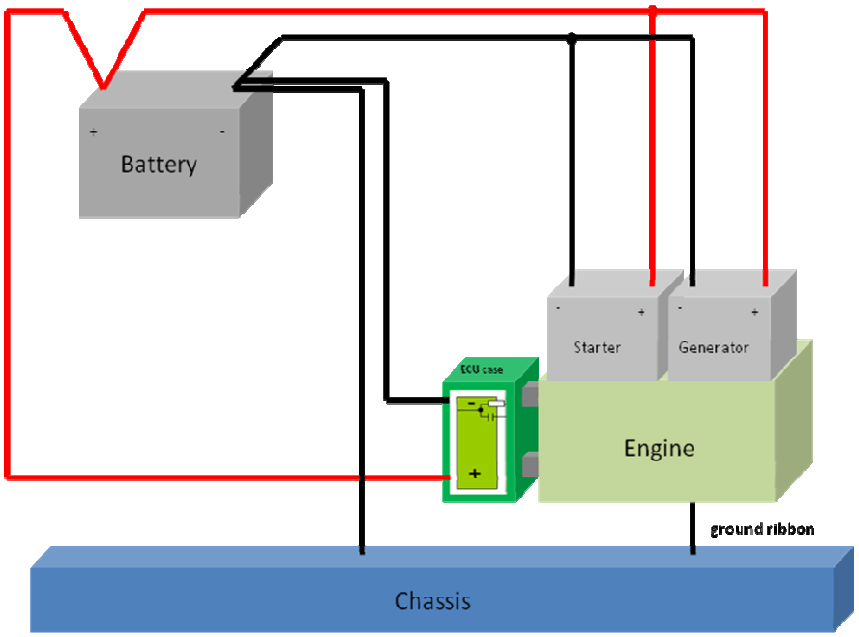


Figure 7 - Recommended ECM power supply / GND connection set-up

The ECM housing is internally not connected to supply GND and it must not be connected separately to ground.

- The following ECM power supply/GND connection set-ups are NOT allowed (see Figure 8):

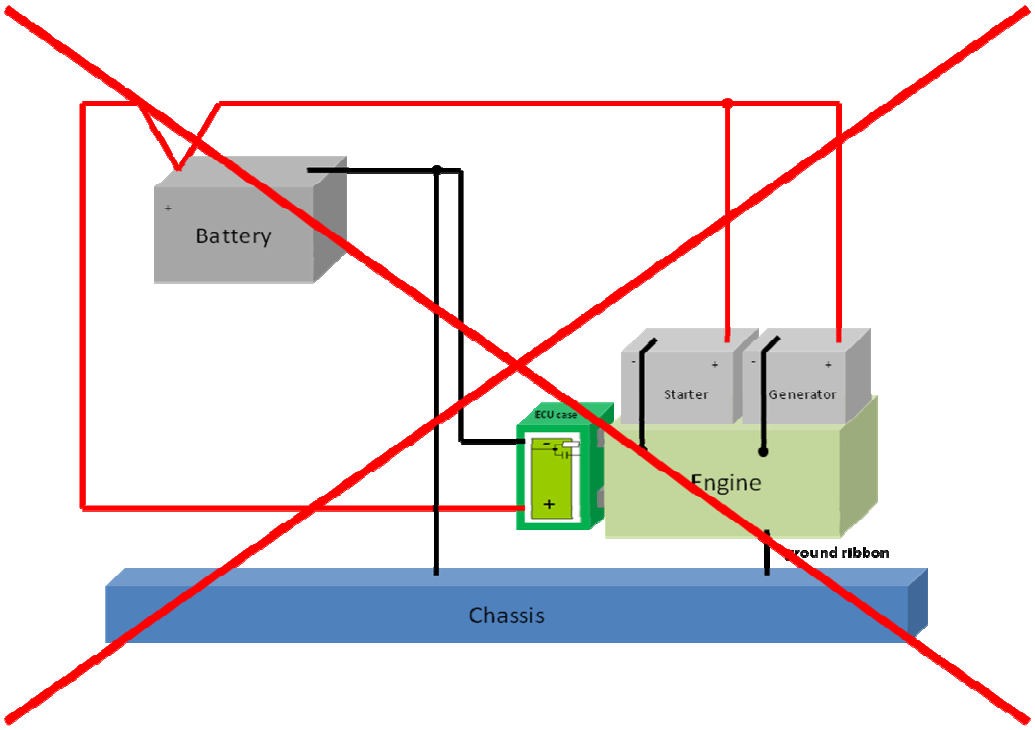


Figure 8 - Wrong ECM power supply / GND connection set-up

Note: this figure shows a set-up having the internal supply GND connection to ECM housing. In this case, if the engine ground connection is faulty, an undesired current can flow through the housing into the ECM PCB.

Important Note: the alternator used to recharge the vehicle battery must be protect against the load-dump, if not provided by FPT.

5.1.2 Temperature limitations

- Operating Temperature: -30 °C to +95 °C.
- The maximum temperature limit of the ECM which must never be exceeded during its active lifetime is 95 °C. This temperature is measured at a location at the bottom plate of the ECM housing which is defined in the following drawing (see Figure 9):

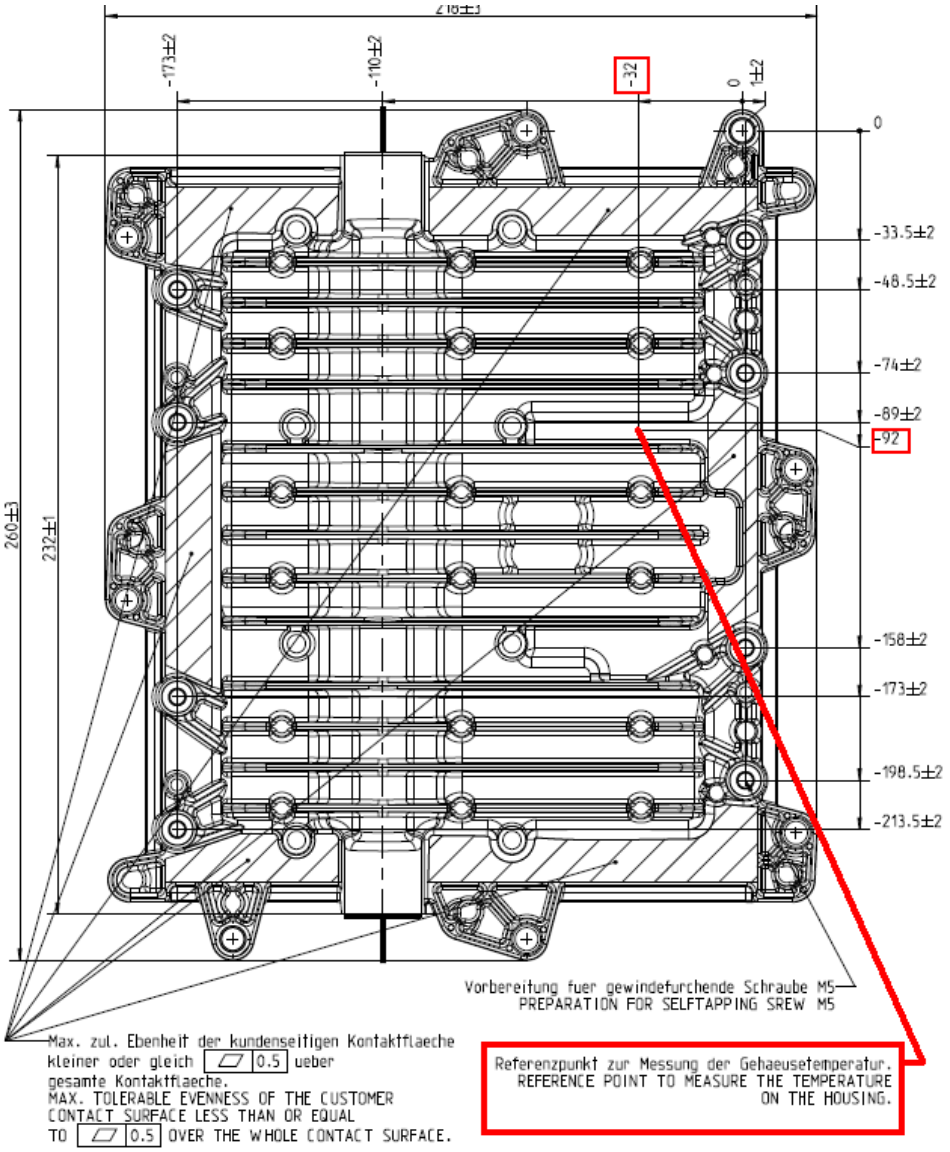


Figure 9 - Bottom view of Bosch ECM

- Storage temperature:
  - Recommended: -30 °C to +40 °C
  - For different values, please contact FPT
- The ECM is mounted on the Engine and it is cooled by the diesel fuel. Actually, it is checked during validation measurement that ECM temperature does not exceed 95 °C limit (measured at bottom plate).
- The best cooling effect is reached when the fuel temperature is in a range from 50 °C to 60 °C. In any case, the fuel temperature must not exceed 75 °C.

5.1.3 Handling

- Touching of connector pins is not allowed.
- Opening of ECM housing is not allowed.
- It has to be assured that no water can infiltrate through the wiring harness into the connector or the ECM respectively. This can be achieved by using a Siphon or a breathing hole (see Figure 10):

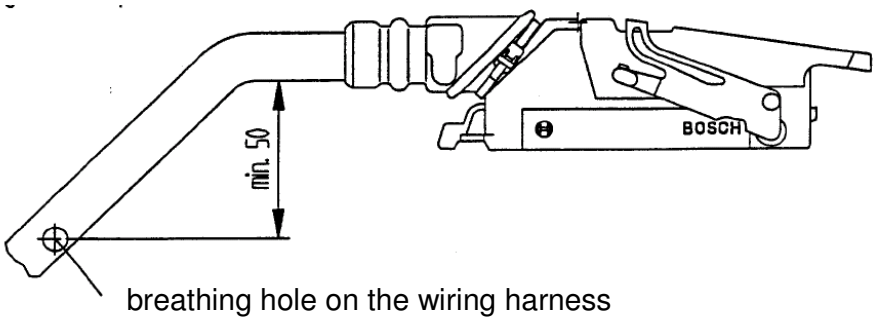


Figure 10 - Bosch ECM Connector

5.1.4 Type of protection

- The ECM has the Standard DIN 40 050 and ISO 20653 protection.
- In addition to this, it has IPX4K/6K/9K protections against splash water at high pressures.

5.1.5 Coating of the Electronic Control Unit

The coating process has to be discussed with FPT. If the Electronic Control Module is coated, it must be ensured, that the Pressure Compensation Element (PCE), the connector and the label are free of coating.

5.2 Wiring Harness Architecture

The ECM input / output can be divided into Engine and Vehicle Connectors (see Figure 11).
The wiring harness, carrying signals from these sections to the other components, is composed by one/two sections:
Engine Connector:

- One section from the ECM connector to the engine components.

Vehicle Connector:

- One section from the ECM connector to some not engine mounted components (filter heaters, water in fuel sensor etc.)

- One section from the ECM connector to the Engine Interface Box (Vehicle Can included)

For further details see
Figure 11.

For an overview of the ECM I/O pin used please refer to the documentation in Appendix D.

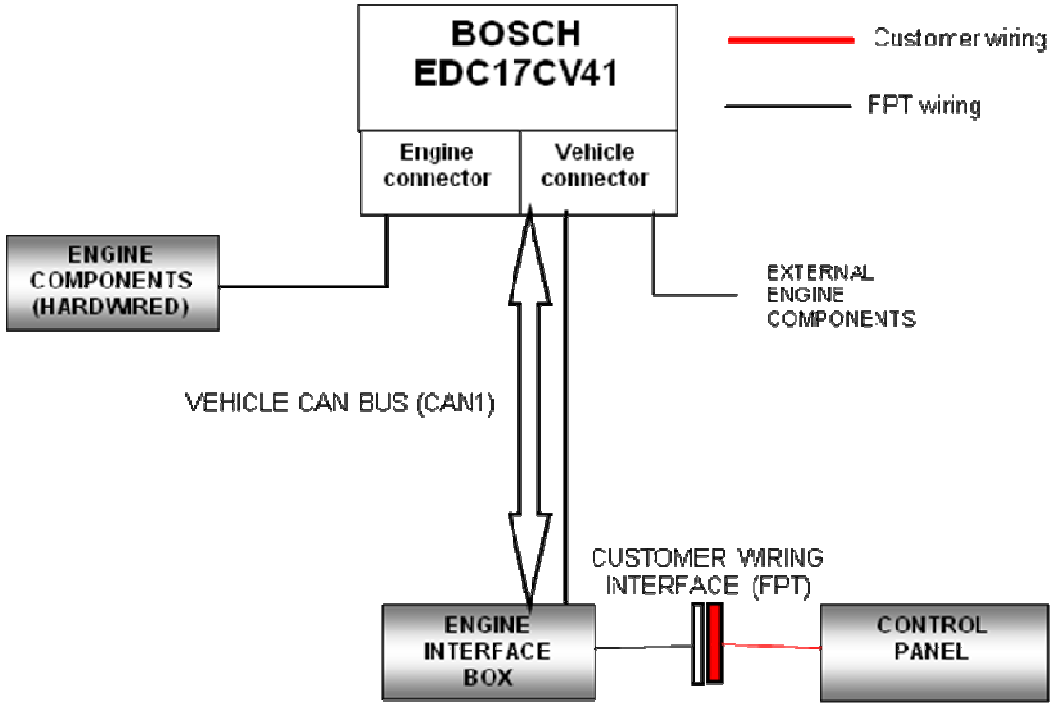


Figure 11 – Wiring Harness Architecture

5.3 ECM Communication

In Figure 12 is shown the Power Generator architecture scheme:

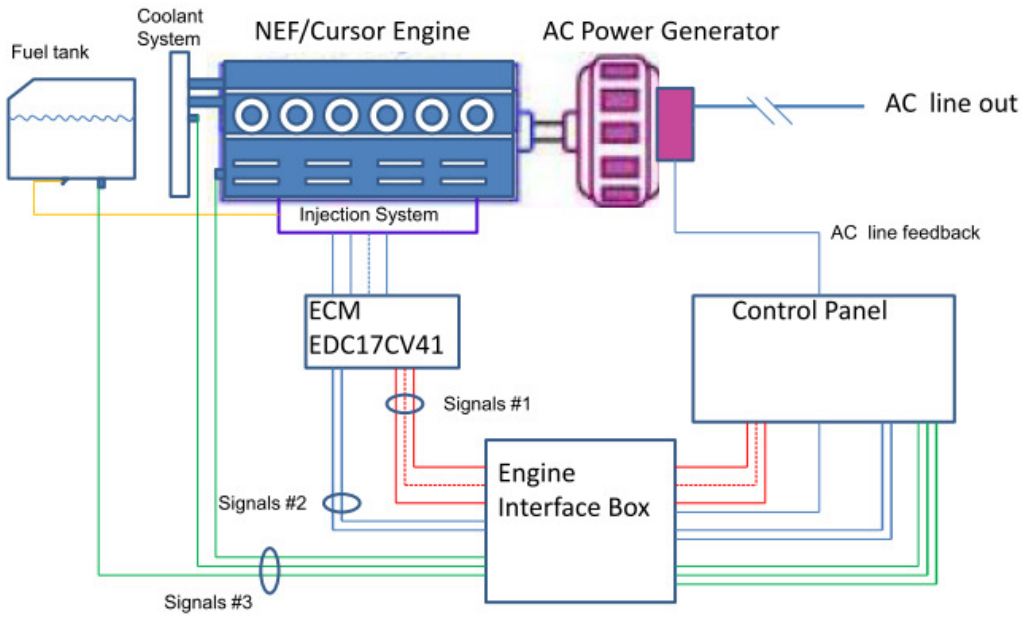


Figure 12 - Power Generator architecture

Signals #1 group: contains 3 type of signals:

- Signals toward ECM that are driven by the Engine Interface Box, as the MSS circuitry for the engine speed selection.
- Signals toward the Engine Interface Box driven by the ECM, as the starter relay , the diagnostic lamp etc.
- Signals toward the Control Panel driven by the ECM, as the cold start lamp. These signals are carried through the Engine Interface Box for harness simplification.

Signals #2 group : contains the J1939 Vehicle Can line of the ECM used by both the Engine Interface Box and the Control Panel.

Signals #3 group : contains signals coming from additional sensors on the fuel tank (low fuel level), the coolant system (high coolant temperature) and the engine (low oil

pressure). These signals are carried to the Control Panel for the alarm condition display and can be used in alternative to the same information (except fuel level) provided by the ECM on the J1939 Can line.

The Engine Interface Box is an electronic equipment connected between the power generator Control Panel and the engine ECM (EDC17CV41). Its purpose is to manage the following functions of the Power Generator:

- 50Hz or 60Hz generator operation, achieved through the selection of two fixed engine speed setpoint, respectively 1500 rpm and 1800 rpm. This is achieved through the ECM MSS input or the vehicle Can line.

- Display engine faults or alarms through a diagnostic lamp.

- Provide actuation relays for some engine functions (engine starter, fuel heater) and dummy loads if one or more of them are not mounted (for example if the engine has not the air heater system). This prevent the ECM to detect faults in case of missing electrical loads on its output.

The engine speed set point can be achieved in two ways:

1. Via MSS through a specific setup of the Engine Interface Box.

2. Via TSC1-PE/TE messages on the vehicle CAN sent to the ECM by the Control Panel.

The selection between the two configurations does not require a change in dataset.
CAN line is in fact configured in dataset as active by default; to get this behaviour and in order to find a compromise between logistic needs and eventual error detection by the power generator user, the ECM error relative to the timeout of CAN messages (TSC1-PE or TSC1-TE) required to control engine speed is calibrated such in a way that diagnostic lamp is not switched on and no error is stored in the ECM fault memory.

With the above described ECM dataset calibration, two possible situations can occur:

- **Configurations without CAN line:** if no errors are active, diagnostic lamp is switched off and no TSC1 timeout error is present in the ECM fault memory. ~~As for this issue, a note could be added in the Genset user manual to notify that the presence of these two error in the ECM fault memory doesn't stand for a real error.~~

- **Configurations with CAN line:** to keep the CAN interface compliant with the Tier4A family, two CAN messages (TSC1-PE and TSC1-TE) are expected to be used, one as alternative to the other, to set and control the desired engine speed. Although this flexibility, no TSC1 timeout failure will be active into the ECM failure memory. In the event of an actual timeout of the TSC1 message used to control the engine speed, diagnostic lamp remains switched off, but Genset user can however detect CAN fault because engine speed drops down to the selected engine speed setpoint.

As shown in Figure 13, there are in total two CAN lines available on the ECM, which can be considered Standard CAN line or Special Purpose CAN line.

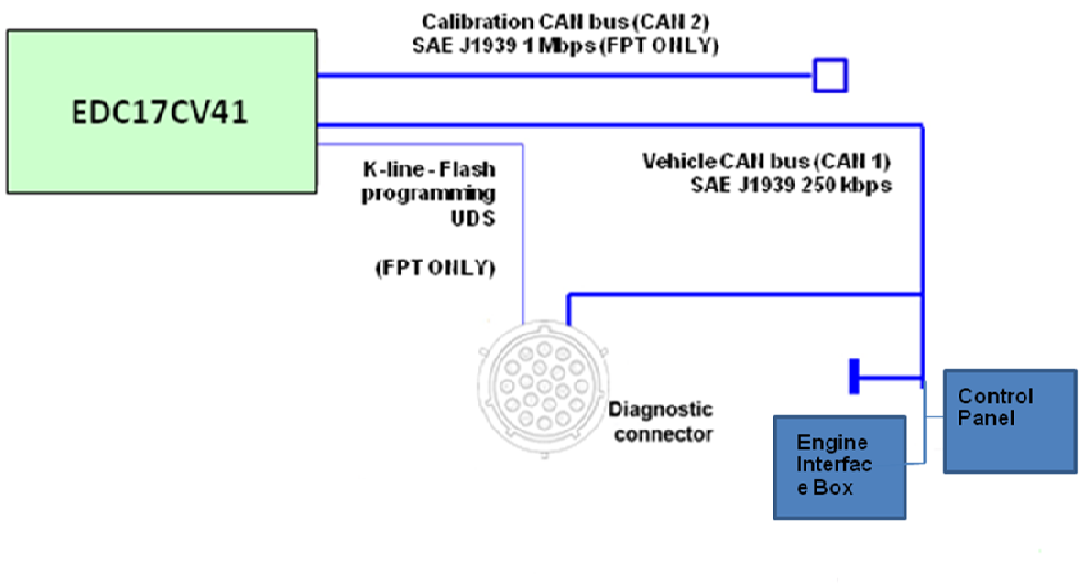


Figure 13 - CAN network

5.3.1 Standard CAN line

Called VEHICLE CAN, it is used to interface with other ECUs and Vehicle Control Module. This CAN bus is compliant to SAE J1939 standard.
Main characteristics are:

- Bit rate: 250 kbit/s
- Configuration: extended 29 bits identifier according to CAN 2.0B
- A CAN termination resistance of 120 Ohm is inside the ECM; the other one must be situated at the other end of the CAN bus.

In order to avoid CAN interface malfunctions, the connections between the various ECUs should be implemented by taking into account the guideline described in Appendix C and [B.1].

For Vehicle CAN matrix, see Appendix B.

5.3.2 Special purpose CAN line (FPT ONLY)

For FPT only special operations (calibration verification/changes) on the ECM vehicle connector is available a third CAN line with following characteristics:

- Bit rate: 1 Mbit/s
- Configuration: extended 29 bits identifier according to CAN 2.0B

A K-line connection is available on the diagnostic connector (see Figure 13) and is used only for FPT EOL purposes. It is not available for the engine customer.

6. ELECTRICAL BALANCE FOR FPT COMPONENTS

Engine Control System Components		I *average max*	I *peak*
ECM	Low idle	-	32 A (12/24V) *
	High idle	-	35 A (12/24V) *
Grid heater	12 V System	180÷240 A	180÷280 A
	24 V System	120÷150 A	160÷290 A
Fuel Filter Heater	12 V System	25 A	
	24 V System	12.5 A	

Table 1 - Electrical Balance For FPT Components

* **Remark**: Peak current absorbed by ECM is calculated considering the worst case (anyhow a 30 A fuse is present), but average current could be far lower in normal conditions.

7. VEHICLE FUNCTIONS

7.1 Engine Torque and Speed Management

For engine controlling purposes the following TSC1 message is used (for details refer to [A.1]):

- TSC1-PE/TE is used for engine speed control; this message is generated by the group controller. A single isochronous governor is available (for details, please refer to [A.1])

For safety reasons, TSC1 implements a counter and a checksum at byte 8 (Message Counter / Checksum).

The **Message Counter** is used to detect situations where the sender repeats the same frame all the time due to malfunction. The receiver of the information may use the counter parameter to detect this situation. The sender will increase the message counter in every cycle. The message counter will count from 0 to 7 and then wraps.

The **Message Checksum** is used to verify the signal path from the sender to the receiver.
The Message Checksum is calculated using the first 7 data bytes, the message counter and the bytes of the message identifier. It is calculated as follows:

- Checksum = (Byte1 + Byte2 + Byte3 + Byte4 + Byte5 + Byte6 + Byte7 + message counter & 0x0F + message ID low byte + message ID mid low byte + message ID mid high byte + message ID high byte)
- Message Checksum = (((Checksum >> 6) & 0x03) + (Checksum >>3) + Checksum) & 0x07

The Message Counter and Checksum should be implemented in VCM as soon as possible, even if only temporarily a default value can be acceptable (to set the byte 8 for Message Counter/ Checksum to FFh).

In case of:

- Checksum failed, the affected TSC1 will be considered by engine control module as DISABLED and the relevant demand, engine speed control, has not effect.
- Message Counter failed, after 5 consecutive message counter with the same value, the affected TSC1 will be considered by engine control module as DISABLED and the relevant demand, engine speed control, has not effect.
- CAN fault, engine speed moves to the engine speed setpoint corresponding to the MSS position selected (for details, please refer to paragraph 7.1.1).

7.1.1 Multiple state switch

MSS function of the EDC is used to select the operating frequency of the Power Generator. In details, MSS can be switched on three available positions:

1. position 0: engine at low-idle (normally 800rpm); in this position, engine speed can be set by external CAN request via TSC1-PE/TE
2. position 1: engine at 1500rpm (for 50Hz power generator) isochronous
3. position 2: engine at 1800rpm (for 60Hz power generator) isochronous

Following is the schematic of the circuit connected to the MSS input on the EDC, for the engine speed selection:

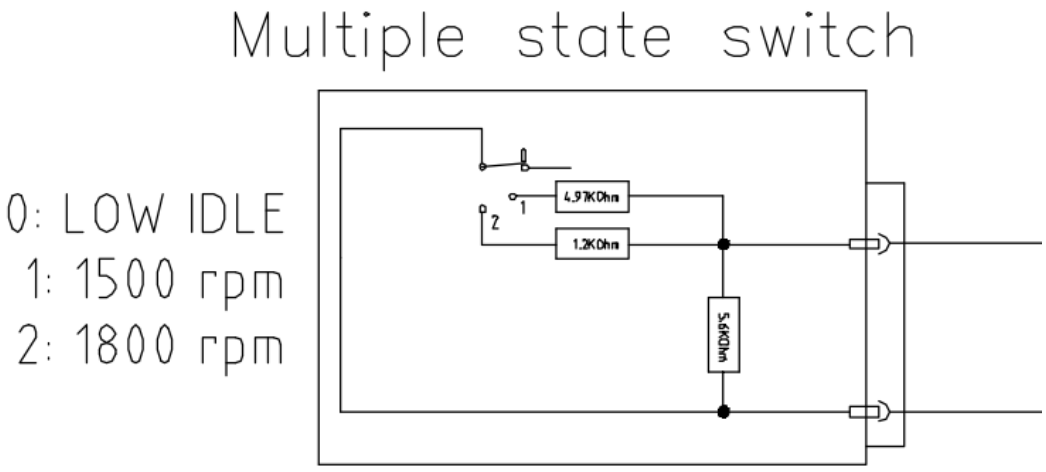


Figure 14 - MSS circuit for engine speed selection

The position is set at Power Generator installation, through proper dip-switches available inside the Engine Interface Box. It's recommended do not change the dip-switch configuration with the engine running, because this cause an immediate change in the engine speed and, thus, in the frequency of the generated AC.

In all the three positions of MSS, if an external engine speed request via CAN becomes active, it has higher priority on the engine speed setpoint selected with MSS.

7.2 High-Idle setting

High-Idle speed is the maximum engine speed allowed in order to avoid engine over speed.
The normal working condition for Power Generator application is engine speed control at 1500rpm or 1800rpm (for 50Hz and 60Hz respectively); low-idle setting has a minor influence; however, if MSS is in position 0 and no engine speed control via TSC1 CAN message is active, engine speed is set at low-idle speed.
High idle setting is useful only to protect the engine and the power generator in which the engine is mounted in the event of external CAN control; high-idle setpoint values, depending on the engine type and displacement, they are set according to the following table:

	F2C (CURSOR 9)	F3B (CURSOR 13)	F4A (NEF)

High-Idle SetPoint	1980 rpm	1980 rpm	1980 rpm

7.3 Water in Fuel Switch

The Water in Fuel Switch, installed in the fuel pre-filter, indicates the presence of water in the fuel, over a fixed threshold dependent from the sensor physical construction.

It is directly hardwired to the ECM vehicular connector (see Appendix D for connections). The state of the sensor is transmitted via CAN on WFI message (for details, please refer to [A.1]).

7.4 Engine Electrical heaters

7.4.1 Fuel Filter and pre-filter Heating

Fuel filter and pre-filter heaters are used to limit the forming of paraffin wax (in case of low temperatures) inside the filtrating element which can stop the fuel flow to the engine.

Fuel filter and pre-filter heating is activated by ECM if fuel temperature is lower than *FuelTHeatThr*Min (and battery voltage is in a proper range) and deactivated if fuel temperature exceeds FuelTHeatThrMax.

In order to improve the engine cranking performances, the heaters are switched off during the cranking phase.

Fuel filter/pre-filter heating relay is directly hardwired to ECM (see Appendix D for connections).

If for some applications fuel filter/pre-filter heating relay is not connected, the customer takes care of connect to EDC17CV41 an equivalent impedance (R = 68Ω ± 10% and P=5W for 12V system, R=120Ω ± 10% and P=5W for 24V system) to simulate the relay presence and to avoid diagnostic errors from the ECM (see Appendix D for connections).

7.4.2 Grid Heater

Grid heater is used in diesel applications that need a quick and reliable start in cold conditions.

Grid heater is installed in the intake manifold and pre-heat the combustion air to the required temperature for ignition of fuel.

It is switched on by ECM according to engine speed, ambient temperature and intake temperature. Moreover, in its activation the battery status is taken into account.

When the grid heater is on, the ECM can command on also a hardwired dedicated lamp (see paragraph 7.5.6) available in the Control Panel.

Grid heater is enabled in dataset and its status is also available via CAN. If for some applications grid heater relay is not connected, the customer takes care of connect to

EDC17CV41 an equivalent impedance (R=68Ω ± 10% and P=5W for 12V system, R=120Ω ± 10% and P=5W for 24V system) to simulate the relay presence and to avoid diagnostic errors from the ECM (see Appendix D for connections).

7.5 Lamps, Alarms and gauges

7.5.1 Engine speed output

Output signal enabled with standard square wave at 6 pulse/rpm, 50% duty-cycle constant. The Engine Speed is also available on the EEC1 CAN message.

7.5.2 High Engine Coolant Temperature Lamp

This lamp warns that Engine coolant temperature exceeded higher limit threshold.
Not hardwired to EDC. Information via CAN by EDC2BC message.
Optionally, if required by the Customer, ECM can also shut down the engine in order to protect it against over-heating.

7.5.3 Low Engine Oil Pressure Lamp

This lamp warns that Engine oil pressure is under lower limit threshold.
Not hardwired to EDC. Information via CAN by EDC2BC message.
Optionally, if required by the Customer, ECM can also shut down the engine in order to protect it against low pressure.

7.5.4 High Engine Oil Temperature Lamp

This lamp warns that Engine oil temperature exceeded higher limit threshold.
Not hardwired to EDC. Information via CAN by EDC2BC message (see [A.1]).

7.5.5 Diagnostic Lamp

The diagnostic lamp can be driven by the ECM through an hardwired connection (Appendix D) or managed via CAN through the DM1 message (see [A.1]). It is mounted into the Engine Interface Box
If the hardwired diagnostic lamp is not used, a dummy resistor (R=1.2 kΩ ± 10%) must be connected to the ECM output, to avoid ghost errors.

7.5.6 Cold Start Lamp

Cold start lamp is an hardwired to EDC lamp, indicating, when switched on, that you have to wait until pre-heating cycle is completed before engine start (see paragraph 7.4.2).

Cold start lamp is enabled in dataset; its status is available via CAN on SHUTDN message (see [A.1]). It is available (if used) in the Control Panel.
If the hardwired cold start lamp is not used, a dummy resistor (R=1.2 kΩ ± 10%) must be connected to avoid diagnostic errors from the ECM.
.

8. DIAGNOSIS

Diagnosis of the ECM and engine malfunctions can be executed in two ways, called on board and off board diagnosis.

8.1 On board diagnosis

The diagnostic information is sent by the ECM directly on the CAN 1 bus through specific messages according to SAE J1939-73 (DM1, DM2, DM3, DM11), mainly intended for diagnostic information to Instrument Cluster.

8.2 Off board diagnosis

Diagnostic information can also be requested to the ECM via the diagnostic connector depicted in Figure 13 with a diagnostic tool PT-BOX, that can send queries to the ECM using the diagnostic protocol UDS (ISO 14229-1 and ISO 15765-2), on CAN 1, and obtain information regarding the engine status, operating conditions, and engine sensors/actuators malfunctions.
ECM programming also possible via K-line (ISO 9141).

APPENDIX A - CALIBRATION SUMMARY AND HINT

The values shown in the following table are indicative and can be changed according to Application Department.

Function	Parameter	Value	EOL programming
Limp home engine speed	*LimpHome*	1200 rpm	No
Minimum fuel temperature threshold for fuel pre-filter heating control	*FuelTHeatThrMin*	0 °C	No
Maximum fuel temperature threshold for fuel pre-filter heating control	*FuelTHeatThrMax*	5 °C	No
Engine Speed set point for PTO 0	*Pto0SetEngSpd*	900 rpm	No
Engine Speed set point for PTO 1	*Pto1SetEngSpd*	1500 rpm	No
Engine Speed set point for PTO 2	*Pto2SetEngSpd*	1800 rpm	No
Engine Speed set point for PTO 3	*Pto3SetEngSpd*	900 rpm	No
PTO 0 Configuration	*Pto0ModeCfg*	137	No
PTO 1 Configuration	*Pto1ModeCfg*	137	No
PTO 2 Configuration	*Pto2ModeCfg*	137	No
PTO 3 Configuration	*Pto3ModeCfg*	137	No

Table 2 – Calibration values

APPENDIX B - CAN MATRIX

Sent messages (vehicle CAN Bus, pins 1.46 - 1.47)

	Message name	Cycle Time [ms]	BUS LOAD [%] Worst Case
Messages transmitted by ECM	EEC1	10	6
	EEC2	50	1.2
	EEC3	50	1.2
	EC1	5000(1)	0.012
	ET1	1000	0.06
	AMB	1000	0.06
	IC1	500	0.12
	EFL/P1	500	0.12
	EFS	500	0.12
	LFE	100	0.6
	LFC	On req	-
	VEP	1000	0.06
	SHUTDN	1000	0.06
	HOURS	On req	-
	EDC2BC	50	1.2
	DM1	1000	0.06
	DM2	On req	-
	Diagnostic Message Ack	On req	-
	STOD	On req	-
	SOFT	On req	-
	DIS	On req	-
	WFI	1000	0.06
	AAI	1000	0.06
	UDS Response	On req	-

Table 3 – Transmitted messages

(1) or in case of a change of speed and/or torque greater than +/-10%.

Received messages (vehicle CAN Bus, pins 1.46 - 1.47)

	Message name	Cycle Time [ms]	BUS LOAD [%] Worst Case
Messages received by ECM	TSC1-PE (1)	10	6
	TSC1-TE (1)	10	6
	PGN Request	On req	-
	UDS Request	On req	-

Table 4 – Received messages

(1) when active, else 200 ms

APPENDIX C - CAN INTERFACE AND FUNCTIONALITY

Main cable parameters are:

- Cable type: twisted shielded cable
- Specific resistance: 25 mOhm/m
- Cable impedance (R_L): 120 Ohm

Referring to to the network topology example in Figure 15 , the connection wiring length are:

- Max bus lenght (L): 40 m
- Max cable stub length (S): 1 m
- Min node distance (d): 0.1 m
- Min distance from R_L (d_0): 0 m

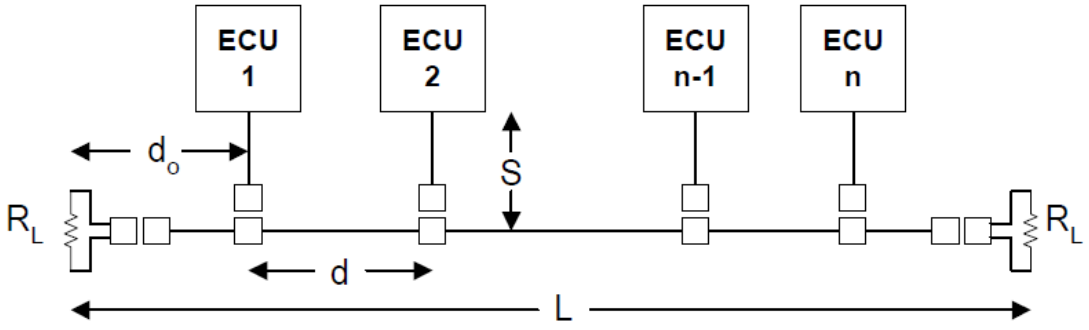


Figure 15 - CAN Network topology

For other network topologies refer to [B.1].

APPENDIX D - EDC17CV41 COMPONENT INTERFACE AND PIN LIST

NEF Engines		
Component	**Pin/Signal**	**Note**
Vehicle interface		
Terminal K15	1.69 I_S_T15	Key switched supply
Terminal K50	1.63 I_S_T50	Start switch term.50 input signal
Vbat Supply	1.01, 1.25, 1.26, 1.49, 1.73 V_V_BAT	ECM Positive Supply
Vbat Ground	1.03, 1.05, 1.28, 1.52, 1.75 G_G_BAT	ECM Ground Return
CAN 1 interface	1.47 B_D_CANL0 1.46 B_D_CANH0	Vehicle CAN
CAN 2 interface	1.95 B_D_CANL1 1.71 B_D_CANH1	Calibration CAN (FPT only)
ISO-K interface	1.70 B_D_ISOK	K-line (FPT only)
Fuel filter/pre-filter heating relay	1.27 O_V_RH31 1.61 O_S_RL20	Vbat voltage from heater
Grid heater relay	1.83 O_S_RL22 1.74 O_V_RH21 (V2)	Vbat voltage from heater
Diagnostic LED	1.77 O_S_RH07 1.51 G_G_RL07	Ground from lamp
Water In Fuel Switch	1.13 I_S_DIG01 1.43 G_R_AN14	Ground from sensor
Engine speed output	1.34 O_F_ENGN	6 Pulse/Rev. open-collector Signal
Starter relay	1.19 O_S_RH04 1.88 O_S_RL21	
Multi State Switch	1.41 I_A_AN26 1.43 G_R_AN14	
Cold Start LED	1.06 O_S_RL25 1.74 O_V_RH21	
Engine interface		
Boost pressure and temperature sensor	2.90 G_R_AN05 2.07 V_V_5VSS1A 2.37 I_A_AN16 2.86 I_A_AN05	

NEF Engines		
Component	**Pin/Signal**	**Note**
Coolant temperature sensor	2.39 I_A_AN17 2.59 G_R_AN18	
Fuel temperature sensor	2.12 I_A_AN18 2.59 G_R_AN18	
Oil pressure and temperature sensor	2.06 G_R_DF03 2.35 I_A_AN01 2.13 I_A_AN15 2.31 V_V_5VSS1F	
Increment speed sensor (crankshaft)	2.66 I_F_CRSPOS 2.65 I_F_CRSNEG 2.69 SHIELD	
Segment speed sensor (camshaft)	2.68 I_F_CASPOS 2.67 G_R_CASNEG 2.69 SHIELD	
Rail pressure sensor	2.60 G_R_RAILPS 2.11 V_V_5VSS3B 2.36 I_A_RAILPS	
Cylinder1 Injector Cylinder2 Injector Cylinder3 Injector Cylinder4 Injector	2.49 O_P_SVH11 2.73 O_P_SVL11 2.26 O_P_SVH22 2.02 O_P_SVL22 2.25 O_P_SVH21 2.01 O_P_SVL21 2.50 O_P_SVH12 2.74 O_P_SVL12	for NEF 4 cyl engine only
Cylinder1 Injector Cylinder2 Injector Cylinder3 Injector Cylinder4 Injector Cylinder5 Injector Cylinder6 Injector	2.49 O_P_SVH11 2.73 O_P_SVL11 2.51 O_P_SVH13 2.75 O_P_SVL13 2.50 O_P_SVH12 2.74 O_P_SVL12 2.27 O_P_SVH23 2.03 O_P_SVL23 2.25 O_P_SVH21 2.01 O_P_SVL21 2.26 O_P_SVH22 2.02 O_P_SVL22	for NEF 6 cyl engine only

Table 5 - I/O interface for EDC17CV41 – NEF Engines

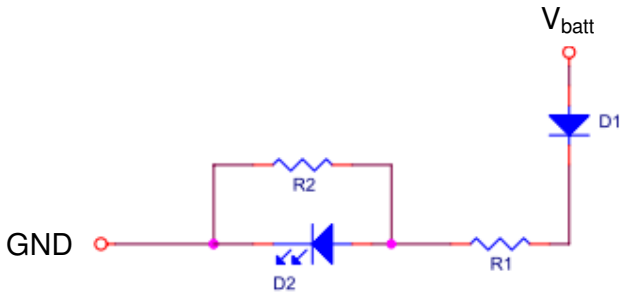
CURSOR Engines		
Component	**Pin/Signal**	**Note**
Vehicle interface		
Terminal K15	1.69 I_S_T15	Key switched supply
Terminal K50	1.63 I_S_T50	Start switch term.50 input signal
Vbat Supply	1.01, 1.25, 1.26, 1.49, 1.73 V_V_BAT	ECM Positive Supply
Vbat Ground	1.03, 1.05, 1.28, 1.52, 1.75 G_G_BAT	ECM Ground Return
CAN 1 interface	1.47 B_D_CANL0 1.46 B_D_CANH0	Vehicle CAN
CAN 2 interface	1.95 B_D_CANL1 1.71 B_D_CANH1	Calibration CAN (FPT ONLY)
ISO-K interface	1.70 B_D_ISOK	K-line (FPT ONLY)
Fuel filter/pre-filter heating relay	1.27 O_V_RH31 1.61 O_S_RL20	Vbat voltage from heater
Grid heater relay	1.83 O_S_RL22 1.74 O_V_RH21 (V2)	Vbat voltage from heater
Diagnostic Lamp	1.77 O_S_RH07 1.51 G_G_RL07	Ground from lamp
Water In Fuel Switch	1.13 I_S_DIG01 1.43 G_R_AN14	Ground from sensor
Engine speed output	1.34 O_F_ENGN	6 Pulse/Rev. open-collector Signal
Starter relay	1.19 O_S_RH04 1.88 O_S_RL21	
Multi State Switch	1.41 I_A_AN26 1.43 G_R_AN14	
Cold Start LED	1.06 O_S_RL25 1.74 O_V_RH21	
Engine interface		
Boost pressure and temperature sensor	2.90 G_R_AN05 2.07 V_V_5VSS1A 2.37 I_A_AN16 2.86 I_A_AN05	
Coolant temperature sensor	2.39 I_A_AN17 2.59 G_R_AN18	
Fuel temperature sensor	2.12 I_A_AN18 2.59 G_R_AN18	
Oil pressure and temperature sensor	2.06 G_R_DF03 2.35 I_A_AN01 2.13 I_A_AN15 2.31 V_V_5VSS1F	
Increment speed sensor (crankshaft)	2.66 I_F_CRSPOS 2.65 I_F_CRSNEG 2.69 SHIELD	

CURSOR Engines		
Component	**Pin/Signal**	**Note**
Segment speed sensor (camshaft)	2.68 I_F_CASPOS 2.67 G_R_CASNEG 2.69 SHIELD	
Rail pressure sensor	2.60 G_R_RAILPS 2.11 V_V_5VSS3B 2.36 I_A_RAILPS	Analog sensor
Cylinder1 Injector Cylinder2 Injector Cylinder3 Injector Cylinder4 Injector Cylinder5 Injector Cylinder6 Injector	2.49 O_P_SVH11 2.73 O_P_SVL11 2.50 O_P_SVH12 2.74 O_P_SVL12 2.51 O_P_SVH13 2.75 O_P_SVL13 2.25 O_P_SVH21 2.01 O_P_SVL21 2.27 O_P_SVH23 2.03 O_P_SVL23 2.26 O_P_SVH22 2.02 O_P_SVL22	Dedicated high voltage drivers
Fuel Metering Unit (ZME)	2.58 O_V_MEU 2.83 O_T_MEU	
Crankcase Pressure Sensor	2.32 V_V_5VSS1E 2.42 G_R_AN20 2.61 I_A_AN03	Analog sensor
Exhaust gas absolute pressure sensor	2.33 V_V5VSS2B 2.16 I_A_AN02 2.19 G_R_AN02	Analog sensor
Fuel Filter Clogging Switch	2.78 G_R_DF02 2.79 I_F_DF02	Fixed threshold pressure switch
Fuel Pre-Filter Clogging Pressure Sensor (C13 ONLY)	2.19 G_R_AN02 2.08 V_V_5VSS2C 2.62 I_A_AN07	Analog sensor

Table 6 - I/O interface for EDC17CV41 – CURSOR Engines

APPENDIX E – LED CONNECTION CIRCUIT

In the following scheme is shown the circuit for connecting LED lamps to the ECM outputs, for 12V and 24V applications.



In case of diagnostic lamp:

- V_{batt} is connected to pin 1.77
- GND is connected to pin 1.51

Parameter	Condition	Min	Typ	Max
Operating Range	Ta-40℃ to +85℃	9V	28.8V	36V
R1			1.2 kΩ	
R2			1.2 kΩ	
Resistor Tolerance				5%
Note: The Diode D1 shall be inserted to decouple all light indicators				

Table 7 - Component values for 12/24V application